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Matrix Folding Theory, Entropy Stacking, and Adsorption Isotherms for Atomistic Simulation

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Abstract

This report documents the theoretical foundations for polymer folding classification, entropy–enthalpy stacking competition, and surface adsorption modelling as they apply to the VSEPR-SIM atomistic simulation platform. We treat RNA-type secondary structure enumeration through matrix methods, define the triangular truncation scheme for planar diagram selection, derive the entropy stacking partition function, and connect these to Langmuir and BET adsorption isotherms relevant to organometallic surface chemistry. All derivations are given in forms suitable for direct implementation in the atomistic kernel.

Contents

Part I
Matrix Folding Theory

1 The Dual-Layer Interaction Matrix

1.1 Definition of the $N \times N \times 2$ System

Consider a polymer of N residues (nucleotides, amino acids, or organometallic repeat units). We define two interaction matrices operating simultaneously on the same index space:

Definition 1.1 (Dual-layer interaction matrix)**.** *Let* $\mathbf{H} \in \mathbb{R}^{N\times N}$ *be the* base-pairing *(or ligand-coordination) energy matrix and* $\mathbf{S} \in \mathbb{R}^{N\times N}$ *be the* stacking *(or π-interaction) energy matrix. The composite system is the pair* $(\mathbf{H}, \mathbf{S})$*, which we write as an* $N \times N \times 2$ *tensor:*

$$\mathcal{M}_{ij}^{(\alpha)} = \begin{cases} H_{ij} & \alpha = 1 \quad \textit{(pairing/coordination)} \\ S_{ij} & \alpha = 2 \quad \textit{(stacking/}\pi\textit{-interaction)} \end{cases} \tag{1}$$

where $i, j \in \{1, \ldots, N\}$ *index residues and* $\alpha \in \{1, 2\}$ *selects the interaction layer.*

For the canonical RNA problem, H_{ij} encodes Watson–Crick complementarity (or wobble pairs) and S_{ij} encodes the stacking free energy between adjacent paired bases. For organometallic polymers with bridging ligands (as in the uranium silyl-amide complexes of Compounds 59 and 60), H_{ij} encodes metal–ligand coordination energies and S_{ij} encodes inter-complex stacking or through-space orbital interactions.

1.2 Small-Scale Grids: The 4×4 Case

In practical atomistic simulation, the smallest non-trivial grid that captures both pairing and stacking is $N = 4$. This gives a $4 \times 4 \times 2$ tensor with 32 independent entries (before symmetry reduction).

For a symmetric polymer:

$$H_{ij} = H_{ji}, \qquad S_{ij} = S_{ji} \tag{2}$$

reducing to $2 \times \binom{4}{2} + 2 \times 4 = 20$ independent parameters.

Remark 1.1. *The* 4×4 *grid is not merely a toy. For RNA, the four nucleotides (A, U, G, C) define a* 4×4 *pairing energy matrix. For organometallic coordination, a four-ligand system (e.g., two amido-N, one oxo-bridge, one silyl group) similarly populates a* 4×4 *coordination-energy grid.*

1.3 Matrix Symmetry and Constraint Classes

We classify entries by their physical role:

Entry type	**Condition**	**Physical meaning**
Diagonal ($i = j$)	H_{ii}, S_{ii}	Self-energy, on-site potential
Near-diagonal ($\lvert i - j\rvert = 1$)	$S_{i,i+1}$	Nearest-neighbour stacking
Off-diagonal ($\lvert i - j\rvert > 1$)	H_{ij}	Long-range pairing / coordination

The stacking matrix $\mathbf{S}$ is typically *banded* (only nearest-neighbour and next-nearest-neighbour terms are significant), while the pairing matrix $\mathbf{H}$ is generally *sparse* but not banded.

2 Secondary Structure as Planar Diagrams

2.1 Definitions

Definition 2.1 (Secondary structure)**.** *A secondary structure on* $\{1, \ldots, N\}$ *is a set of pairs* $\sigma = \{(i_1, j_1), \ldots, (i_k, j_k)\}$ *such that:*

1. *Each index appears in at most one pair (matching condition).*
2. *If* $(i, j) \in \sigma$ *with* $i < j$*, then* $j - i > 3$ *(minimum loop length).*
3. *If* $(i, j), (k, l) \in \sigma$ *with* $i < k$*, then either* $j < k$ *or* $i < k < l < j$ *(no crossing; planarity).*

Condition 3 is the *planarity constraint*: when we draw arcs connecting paired indices on a line, no two arcs cross. This is precisely the condition that distinguishes *secondary* from *tertiary* structure (pseudoknots).

2.2 Counting Secondary Structures

The number of secondary structures on N residues (with minimum loop length 1 for simplicity) satisfies the Catalan-like recurrence:

$$s(N) = s(N-1) + \sum_{k=1}^{N-2} s(k-1)\, s(N-k-1) \tag{3}$$

with $s(0) = s(1) = 1$. The asymptotic growth is:

$$s(N) \sim C \cdot N^{-3/2} \cdot \alpha^N, \qquad \alpha \approx 2.618\ldots \tag{4}$$

For the partition function, each structure σ is weighted by its Boltzmann factor:

$$Z = \sum_{\sigma} \exp\Big(-\beta\, E(\sigma)\Big), \qquad E(\sigma) = \sum_{(i,j)\in\sigma} H_{ij} + \sum_{\substack{(i,j),(i+1,j-1)\\ \in\sigma}} S_{i,i+1} \tag{5}$$

3 Triangular Truncation

3.1 Motivation

The full $N \times N$ interaction matrix contains $O(N^2)$ entries, but the secondary structure constraint restricts us to the upper triangle with the additional planarity filter. *Triangular truncation* formalizes this by zeroing out entries that cannot participate in valid secondary pairings.

Definition 3.1 (Triangular truncation operator)**.** *For a matrix* $\mathbf{A} \in \mathbb{R}^{N\times N}$*, define:*

$$[\mathcal{T}(\mathbf{A})]_{ij} = \begin{cases} A_{ij} & \textit{if } j - i > d_{\min} \\ 0 & \textit{otherwise} \end{cases} \tag{6}$$

where $d_{\min}$ *is the minimum loop length (typically 3–4 for RNA, variable for organometallic chains).*

3.2 Spectral Properties

The truncated matrix $\mathcal{T}(\mathbf{H})$ has well-known spectral properties. For Gaussian random matrices (the "large-N" limit of random polymer theory):

Theorem 3.1 (Wigner semicircle under truncation). *If* $\mathbf{H}$ *is drawn from GOE*(N) *and* $d_{\min} = O(1)$*, then* $\mathcal{T}(\mathbf{H})$ *retains the semicircle law in the* $N \to \infty$ *limit, with spectral radius scaled by* $\sqrt{1 - d_{\min}/N}$*.*

In practice, we are far from the large-N limit ($N = 4$ to $N \sim 100$ for the systems of interest), so the spectral structure must be computed explicitly.

3.3 Implementation in the Atomistic Kernel

The triangular truncation is applied during matrix assembly:

```cpp
// atomistic kernel: triangular truncation
template <int N>
void triangular_truncate(double H[N][N], int d_min) {
    for (int i = 0; i < N; ++i)
        for (int j = 0; j < N; ++j)
            if (std::abs(j - i) <= d_min)
                H[i][j] = 0.0;
}
```

This reduces the effective matrix density from $O(N^2)$ to $O(N^2 - N \cdot d_{\min})$ and eliminates non-physical short-range "pairings" that would violate loop-length constraints.

4 The Large-N Limit and Planar Diagrams

4.1 't Hooft Expansion

In the matrix model formulation, the partition function admits a topological expansion in $1/N^2$:

$$\ln Z = \sum_{g=0}^{\infty} N^{2-2g} F_g \tag{7}$$

where g is the genus of the corresponding surface. The leading term ($g = 0$) corresponds to *planar* diagrams — precisely the secondary structures with no crossing arcs.

Proposition 4.1. *In the large-N limit, only planar (secondary) structures contribute to the free energy at leading order. Pseudoknots (tertiary structures) are suppressed by* $O(1/N^2)$*.*

This provides the mathematical justification for the triangular truncation: at leading order, the truncated matrix captures all physically relevant interactions.

4.2 Connection to Organometallic Chains

For bridged organometallic polymers (e.g., the oxo-bridged diuranium complexes), the "planarity" condition translates to the requirement that bridging ligands connect only non-crossing pairs of metal centres. In the silyl-amide systems (Compounds 59 and 60), the oxo bridge (μ-O) connects two uranium centres without crossing any of the amido-N coordination bonds — a geometrically planar arrangement consistent with the secondary structure formalism.

Part II
Entropy Stacking Thermodynamics

5 The Stacking Free Energy

5.1 Enthalpy–Entropy Competition

When planar molecular units (nucleotide bases, porphyrin rings, organometallic coordination planes) stack, there is a thermodynamic competition:

$$\Delta G_{\text{stack}} = \Delta H_{\text{stack}} - T\Delta S_{\text{stack}} \tag{8}$$
$$\Delta H_{\text{stack}} < 0 \quad \text{(favourable: van der Waals, } \pi\text{–}\pi\text{, dispersion)} \tag{9}$$
$$\Delta S_{\text{stack}} < 0 \quad \text{(unfavourable: ordering reduces configurational freedom)} \tag{10}$$

The net stacking propensity depends on temperature:

$$T^* = \frac{\Delta H_{\text{stack}}}{\Delta S_{\text{stack}}} \tag{11}$$

is the *stacking melting temperature* — above T^*, entropy dominates and the stack disassembles.

5.2 Nearest-Neighbour Model

In the nearest-neighbour approximation, the total stacking energy for a chain of N residues is:

$$E_{\text{stack}} = \sum_{i=1}^{N-1} S_{i,i+1}\,\sigma_i\,\sigma_{i+1} \tag{12}$$

where $\sigma_i \in \{0, 1\}$ indicates whether residue i participates in a stack. The partition function factorizes as a transfer matrix:

$$Z_{\text{stack}} = \operatorname{Tr}\left(\prod_{i=1}^{N-1} \mathbf{T}_i\right), \qquad \mathbf{T}_i = \begin{pmatrix} 1 & 1 \\ 1 & e^{-\beta S_{i,i+1}} \end{pmatrix} \tag{13}$$

5.3 Entropy Contribution per Stack

The configurational entropy of a partially stacked chain with k stacked pairs out of $N-1$ possible:

$$S_{\text{config}}(k) = k_B \ln\binom{N-1}{k} \approx k_B\left[(N-1)H\left(\frac{k}{N-1}\right)\right] \tag{14}$$

where $H(x) = -x\ln x - (1-x)\ln(1-x)$ is the binary entropy function.

The optimal stacking fraction $\phi^* = k^*/(N-1)$ satisfies:

$$\frac{\partial}{\partial\phi}\left[\phi\,\langle S_{i,i+1}\rangle - k_B T\,H(\phi)\right] = 0 \implies \phi^* = \frac{1}{1+\exp\!\big(\beta\,\langle S_{i,i+1}\rangle\big)} \tag{15}$$

which is a Fermi–Dirac distribution over stacking sites.

6 Entropy Stacking in Matrix Ensembles

6.1 Ensemble Stacking (Machine Learning Analogy)

The term "entropy stacking" also appears in ensemble methods, where multiple base models (each represented by a matrix of predictions) are combined using entropy-weighted averaging:

$$\hat{y} = \sum_{m=1}^{M} w_m \, \hat{y}_m, \qquad w_m \propto \exp\bigl(-\lambda \, H_m\bigr) \tag{16}$$

where H_m is the entropy of model m's prediction distribution. Lower-entropy (more confident) models receive higher weight.

Remark 6.1. *This is formally identical to the Boltzmann weighting of molecular configurations if we identify $\lambda \leftrightarrow \beta$ and $H_m \leftrightarrow E_m$. The VSEPR-SIM kernel can therefore reuse the same partition function machinery for both physical stacking calculations and model-ensemble scoring.*

6.2 Implementation: Combined Scoring

The combined health/quality score for a stacked ensemble of M atomistic models:

$$Q = \frac{\sum_m e^{-\lambda H_m} \, q_m}{\sum_m e^{-\lambda H_m}} \tag{17}$$

where q_m is the individual model quality score and H_m is its prediction entropy. This connects directly to the flower health indicator system's weighted deduction model, where each health metric acts as a "model" contributing to the overall score.

Part III

Adsorption Isotherms for Surface Chemistry

7 Langmuir Adsorption

7.1 Single-Layer Model

The Langmuir isotherm assumes monolayer adsorption on a surface with N_s equivalent, non-interacting sites:

$$\theta = \frac{K_L \, C}{1 + K_L \, C} \tag{18}$$

where:

- θ is the fractional surface coverage
- C is the bulk concentration (or partial pressure)
- $K_L = k_a/k_d$ is the Langmuir equilibrium constant (ratio of adsorption to desorption rate constants)

The adsorbed quantity per unit mass:

$$q_t = q_{\max} \, \theta = \frac{q_{\max} \, K_L \, C}{1 + K_L \, C} \tag{19}$$

7.2 Kinetic Derivation

At equilibrium, the rate of adsorption equals the rate of desorption:

$$r_{\text{ads}} = k_a\, C\,(1-\theta) \tag{20}$$
$$r_{\text{des}} = k_d\, \theta \tag{21}$$
$$r_{\text{ads}} = r_{\text{des}} \implies \theta = \frac{K_L\, C}{1 + K_L\, C} \tag{22}$$

The approach to equilibrium follows pseudo-first-order or pseudo-second-order kinetics:

$$\text{Pseudo-first-order:} \quad \frac{dq_t}{dt} = k_1(q_e - q_t) \tag{23}$$
$$\text{Pseudo-second-order:} \quad \frac{dq_t}{dt} = k_2(q_e - q_t)^2 \tag{24}$$

7.3 Linearized Forms for Fitting

Form	**Plot**	**Equation**
Langmuir-1	C/q_t vs. C	$\frac{C}{q_t} = \frac{1}{q_{\max}K_L} + \frac{C}{q_{\max}}$
Langmuir-2	$1/q_t$ vs. $1/C$	$\frac{1}{q_t} = \frac{1}{q_{\max}} + \frac{1}{q_{\max}K_L C}$
Freundlich	$\ln q_t$ vs. $\ln C$	$q_t = K_F\, C^{1/n}$

8 BET Adsorption (Multilayer)

8.1 The BET Equation

The Brunauer–Emmett–Teller (BET) model extends Langmuir to multilayer adsorption. Each adsorbed layer s_i serves as a substrate for the next layer:

$$s_i = C_{\text{BET}}\, x^i\, s_0, \qquad x = \frac{p}{p_0} \tag{25}$$

where:

- s_0 = number of bare surface sites
- s_i = number of sites covered by exactly i layers
- $C_{\text{BET}} = \exp\big[(\varepsilon_1 - \varepsilon_L)/k_B T\big]$ is the BET constant
- ε_1 = first-layer adsorption energy
- ε_L = liquefaction energy (layers ≥ 2)
- $x = p/p_0$ = relative pressure

The total adsorbed volume:

$$V = V_0 \sum_{i=0}^{\infty} i\, s_i = \frac{V_m\, C_{\text{BET}}\, x}{(1-x)\big[1 + (C_{\text{BET}} - 1)\, x\big]} \tag{26}$$

8.2 Derivation via Geometric Series

The total number of occupied layers:

$$N_{\text{total}} = \sum_{i=1}^{\infty} i\, s_i = C_{\text{BET}}\, s_0 \sum_{i=1}^{\infty} i\, x^i = \frac{C_{\text{BET}}\, s_0\, x}{(1-x)^2} \tag{27}$$

The total number of sites (occupied and bare):

$$N_s = s_0 + \sum_{i=1}^{\infty} s_i = s_0 \left(1 + \frac{C_{\text{BET}}\, x}{1-x}\right) = \frac{s_0\big[1 + (C_{\text{BET}} - 1)\, x\big]}{1-x} \tag{28}$$

The ratio yields Equation (**??**).

8.3 Linearized BET Form

Rearranging Equation (**??**):

$$\frac{x}{V(1-x)} = \frac{1}{V_m\, C_{\text{BET}}} + \frac{C_{\text{BET}} - 1}{V_m\, C_{\text{BET}}}\, x \tag{29}$$

A plot of $x/[V(1-x)]$ vs. x gives a straight line with:

$$\text{slope} = \frac{C_{\text{BET}} - 1}{V_m\, C_{\text{BET}}} \tag{30}$$

$$\text{intercept} = \frac{1}{V_m\, C_{\text{BET}}} \tag{31}$$

From which V_m and C_{BET} are extracted.

9 Application to Organometallic Surface Chemistry

9.1 Uranium Coordination Polymers

The diuranium oxo-bridged complexes (Compounds 59 and 60) exhibit surface-like coordination chemistry:

Property	**Compound 59**	**Compound 60**
Metal centres	U—O—U	U—O—U
N-donor ligands	4 (NR, NR)	4 (NR_2)
Si substituents	Bu^t, Me	Me_3Si
Coordination no.	5–6	5–6
Bridging mode	μ-O + CH_2	μ-O

The coordination sites on each uranium centre act as “adsorption sites” in the Langmuir sense: each site binds at most one ligand, and the binding energy depends on the ligand identity and the trans-influence of other coordinated groups.

9.2 Selectivity and Dissociation

For competitive binding (cf. the selectivity data for Na^+, K^+, Ca^{2+}, Mg^{2+}, Fe^{3+}, Co^{2+}, Cu^{2+}, Ni^{2+}, VO^{3-}), the multi-component Langmuir isotherm applies:

$$\theta_j = \frac{K_j\,C_j}{1 + \sum_k K_k\,C_k} \tag{32}$$

The dissociation constant K_d (reported as $K_d = 3.35\,\mathrm{nM}$ for the uranyl system) is the inverse of the Langmuir constant:

$$K_d = \frac{1}{K_L} = \frac{k_d}{k_a} \tag{33}$$

The fraction of bound uranium as a function of total carbonate concentration follows the competitive displacement curve:

$$f_{\mathrm{bound}} = \frac{K_d}{K_d + [\mathrm{CO}_3^{2-}]_{\mathrm{total}}} \tag{34}$$

Part IV
Unified Computational Framework

10 The Complete Simulation Pipeline

The four theoretical components unite into a single simulation workflow:

1. **Define the interaction matrices.** Construct $\mathbf{H}$ (pairing/coordination) and $\mathbf{S}$ (stacking) from atomistic force field parameters or quantum chemical calculations.

2. **Apply triangular truncation.** Zero out entries with $|i-j| \leq d_{\mathrm{min}}$ to enforce the minimum loop-length constraint and restrict to secondary (planar) structures.

3. **Compute the partition function.** Enumerate valid secondary structures (via dynamic programming) and weight each by its Boltzmann factor incorporating both pairing and stacking energies.

4. **Extract thermodynamic observables.** From Z, compute:

   - Free energy: $F = -k_B T \ln Z$
   - Pairing probability: $p_{ij} = \partial \ln Z / \partial(\beta H_{ij})$
   - Stacking fraction: $\langle \phi \rangle = -\partial \ln Z / \partial(\beta \langle S \rangle)$
   - Adsorption isotherms for surface-bound species

5. **Report and visualize.** Generate structured output suitable for research documentation, including isotherms, folding landscapes, and selectivity charts.

11 Data Structures in the Atomistic Kernel

```
// Dual-layer interaction system
struct DualLayerMatrix {
    int                 N;          // residue count
    std::vector<double> pairing;    // N*N, row-major (H)
    std::vector<double> stacking;   // N*N, row-major (S)
    int                 d_min;      // minimum loop length

    double& H(int i, int j) { return pairing[i * N + j]; }
    double& S(int i, int j) { return stacking[i * N + j]; }

    void truncate() {
        for (int i = 0; i < N; ++i)
            for (int j = 0; j < N; ++j)
                if (std::abs(j - i) <= d_min) {
                    H(i, j) = 0.0;
                    S(i, j) = 0.0;
                }
    }
};

// Langmuir isotherm
struct LangmuirSite {
    double K_L;     // equilibrium constant
    double q_max;   // maximum capacity (mg/g)

    double coverage(double C) const {
        return (K_L * C) / (1.0 + K_L * C);
    }
    double capacity(double C) const {
        return q_max * coverage(C);
    }
};

// BET multilayer
struct BETModel {
    double V_m;      // monolayer volume
    double C_BET;    // BET constant

    double volume(double x) const {
        return (V_m * C_BET * x)
             / ((1.0 - x) * (1.0 + (C_BET - 1.0) * x));
    }
};
```

12 Partition Function via Dynamic Programming

The Nussinov-style recursion for the secondary structure partition function with stacking:

$$Z(i,j) = \underbrace{Z(i+1,j)}_{i\text{ unpaired}} + \sum_{\substack{k=i+d_{\min}+1 \\ k\le j}} \underbrace{e^{-\beta H_{ik}}}_{\text{pair }(i,k)} \cdot \underbrace{Z(i+1,k-1)}_{\text{interior}} \cdot \underbrace{Z(k+1,j)}_{\text{exterior}} \tag{35}$$

with stacking bonus applied when consecutive pairs (i, k) and $(i+1, k-1)$ are both present:

$$Z_{\text{stack}}(i,j) = Z(i,j) + \sum_{\substack{(i,k)\in\sigma \\ (i+1,k-1)\in\sigma}} e^{-\beta S_{i,i+1}}\, Z(i+2,k-2)\, Z(k+1,j) \tag{36}$$

```
// Nussinov partition function with stacking
double partition_dp(const DualLayerMatrix& M, double beta) {
    int N = M.N;
    std::vector<std::vector<double>> Z(N, std::vector<double>(N, 1.0));

    for (int len = 2; len <= N; ++len) {
        for (int i = 0; i + len - 1 < N; ++i) {
            int j = i + len - 1;
            Z[i][j] = Z[i + 1][j];  // i unpaired
            for (int k = i + M.d_min + 1; k <= j; ++k) {
                double pair_weight = std::exp(-beta * M.pairing[i * N + k]);
                Z[i][j] += pair_weight * Z[i + 1][k - 1] * Z[k + 1][j];
            }
        }
    }
    return Z[0][N - 1];
}
```

The time complexity is $O(N^3)$ and space is $O(N^2)$, which is tractable for $N \leq 10{,}000$.

13 Summary of Key Equations

System	Key equation
Dual-layer tensor	$\mathcal{M}_{ij}^{(\alpha)} = H_{ij}\,\delta_{\alpha 1} + S_{ij}\,\delta_{\alpha 2}$
Triangular truncation	$[\mathcal{T}(\mathbf{A})]_{ij} = A_{ij}\,\Theta(\lvert i-j\rvert - d_{\min})$
Partition function	$Z = \sum_\sigma \exp\big(-\beta E(\sigma)\big)$
Stacking fraction	$\phi^* = \big[1 + e^{\beta\langle S\rangle}\big]^{-1}$
Langmuir	$\theta = K_L C/(1 + K_L C)$
BET	$V = V_m C x/[(1-x)(1+(C-1)x)]$
Competitive binding	$\theta_j = K_j C_j/(1 + \sum_k K_k C_k)$

A Notation Index

Symbol	Meaning
N	Number of residues / sites
$\mathbf{H}$	Pairing / coordination energy matrix
$\mathbf{S}$	Stacking / π-interaction energy matrix
α	Layer index $\in \{1, 2\}$
$d_{\min}$	Minimum loop length
σ	Secondary structure (set of non-crossing pairs)
β	Inverse temperature $1/(k_B T)$
Z	Partition function
θ	Fractional surface coverage
K_L	Langmuir equilibrium constant
K_d	Dissociation constant ($= 1/K_L$)
C_{BET}	BET constant
V_m	Monolayer volume
x	Relative pressure p/p_0
s_i	Sites with exactly i adsorbed layers
ϕ	Stacking fraction

B Computational Complexity

Algorithm	Time	Space
Triangular truncation	$O(N^2)$	$O(N^2)$
Nussinov partition (DP)	$O(N^3)$	$O(N^2)$
Stacking transfer matrix	$O(N)$	$O(1)$
Langmuir isotherm	$O(1)$	$O(1)$
BET isotherm	$O(1)$	$O(1)$
Competitive multi-site	$O(M)$	$O(M)$